

Day 29 – AI Conversation Flow Design

Zecpath AI System • Implementation & Validation Documentation

Objective

Define how AI interacts dynamically during calls by designing conversation decision logic, handling silence and response issues, defining fallback questions, triggering follow-ups, and implementing polite failure and retry logic.

Day 29 Requirements

Requirement	Status
AI call decision tree	Completed
Silence handling	Completed
Confusion handling	Completed
Repeated-answer handling	Completed
Fallback questions	Completed
Follow-up triggers	Completed
Polite failure and retry logic	Completed

1. Conversation State Schema

The ConversationState dataclass stores the current question, conversation status, answers, issue counters, retry information, fallback question, follow-up status, and the latest error. Key fields include candidate_id, current_question_id, current_question, conversation_status, last_answer, previous_answers, silence_count, confusion_count, repeated_answer_count, retry_count, fallback_question, follow_up_triggered, and last_error.

2. AI Call Decision Tree

The decision tree determines the next conversation action according to the current state. Completed conversations end, failed conversations enter retry handling, and active issue states are routed to their corresponding handlers. Decision priority: completed → failed → silence → confusion → repeated answer → follow-up → continue conversation.

3. Silence Handling

Silence is handled progressively. The first silence waits for a response, the second offers to repeat the question, and further silence moves the conversation to the next question.

4. Confusion & Repeated Answer Handling

For confusion, the AI first clarifies the question, then rephrases it, and eventually moves to the next question. For repeated answers, the AI first requests additional details, then asks for a different example, and finally moves forward.

5. Fallback Questions & Follow-up Triggers

Fallback questions are provided for technical, experience, skills, and general categories. A follow-up trigger marks the state and stores the appropriate fallback question so the conversation flow can ask it.

6. Polite Failure & Retry Logic

Conversation failures are handled politely using a retry limit of two retries. The first two failures request another attempt. When the retry limit is exceeded, the conversation is marked failed and ends with a polite message.

7. Complete Conversation Flow

The conversation_flow module integrates the state schema, decision tree, silence handler, response issue handlers, follow-up handler, and failure/retry handler into a single process_conversation function. The integrated flow supports normal continuation, silence handling, confusion handling, repeated-answer handling, follow-up questions, failure/retry behavior, and completed-conversation termination.

8. Final Validation Testing

The final Day 29 validation test covered seven scenarios: normal conversation, silence handling, confusion handling, repeated answers, follow-up questions, failure and retry, and completed conversation.

Validation Test	Result
Normal Conversation	PASS
Silence Handling	PASS
Confusion Handling	PASS
Repeated Answer Handling	PASS
Follow-up Question	PASS
Failure & Retry	PASS
Completed Conversation	PASS

Final Result: ALL DAY 29 CONVERSATION FLOW TESTS PASSED

Day 29 AI Conversation Flow Validation: SUCCESS

Completed Deliverables

- AI call flow logic
- Conversation state machine
- Error-handling flow design

Day 29 Status: COMPLETE

All 8 implementation and validation steps have been completed successfully.